

Physics question bank for the third formative test

Unit 11: Modern physics		الوحدات التدريبية
TPK01 – TPK02		أكواد المعارف
Unit 11: Explain evidences of modern physics, X- rays and Laser		المعارف
Radiation of Black Body and The Phenomenon Of The Thermal Ion Emission And Photoelectric	TPK01	
Compton Scattering and X- Rays & LASER	TPK02	

Unit 11	Lesson one Lesson two Lesson three	Lesson four Lesson five Lesson six
Knowledge code	TPK01	TPK02

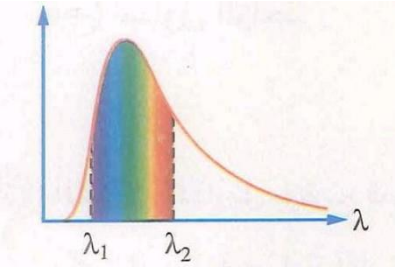
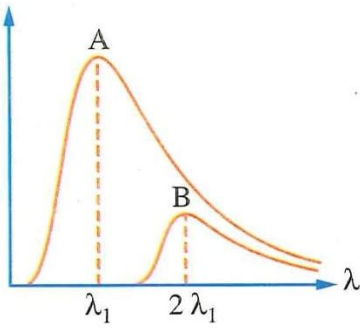
المعرفة	الكود
Radiation of Black Body	TPK01

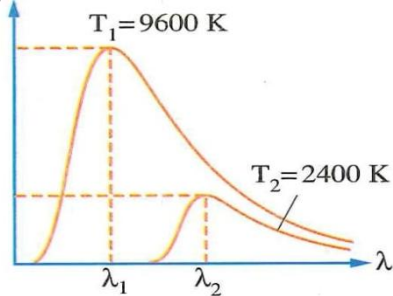
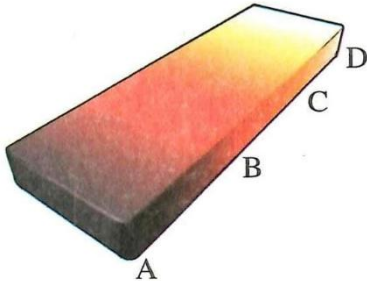
1- Choose the correct answer:

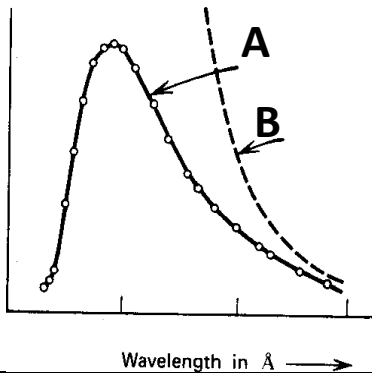
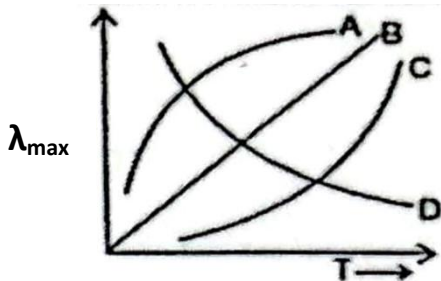
#	Question	Choices	Answer
1	Which of the following electromagnetic waves have the shortest wavelength	a. IR b. X-Rays c. UV d. Visible Light	B
2	The electromagnetic radiation that have the highest frequency and the shortest wavelength.....	a. Radio waves. b. Visible waves. c. Gama rays .	C
3	Due to the classical expectation the intensity of radiation increases by	a. Decreasing wavelength of radiation b. Decreases frequency of radiation c. Increases wavelength of radiation	A
4	the wavelength of a photon is 2.2×10^{-7} m, its energy is given by	a. 1×10^{35} joule b. 9×10^{-19} joule c. 3.3×10^{-27} joule d. 4.4×10^{-32} joule	B

5	Difference between modern physics and quantum physics	a. Modern physics talks about macroscopical model b. Quantum physics talks about macroscopical model c. Both are the same	C
6	Difference between modern physics and classical physics	a. Modern physics talks about macroscopical model b. Classical physics talks about macroscopical model c. Both are the same	B
7	The emitted energy of the sun radiation is ----- visible light.	a. 40% b. 50% c. 20% d. 80%	A
8	The emitted energy of the sun radiation is ----- heat radiation.	a. 40% b. 50% c. 20% d. 80%	B
9	The emitted energy of the lamp radiation is ----- visible light.	a. 40% b. 50% c. 20% d. 80%	C
10	The emitted energy of the lamp radiation is ----- heat radiation.	a. 40% b. 50% c. 20% d. 80%	D
11	The electromagnetic radiation that have the lowest frequency and the tallest wavelength.....	a. Radio waves. b. Visible waves. c. Gama rays .	A
12	It explains all phenomena involved in electronics which is the basis for all modem electronic and communication systems.	a. Classical physics. b. Quantum physics. c. All the above .	B

1 3	The speed of electromagnetic waves equals	a. $3 \times 10^8 \frac{m}{s}$ b. $3 \times 10^{-8} \frac{m}{s}$ c. $4 \times 10^8 \frac{m}{s}$	A															
1 4	The speed of electromagnetic waves is calculated by the relation	a. $C = \lambda v$ b. $C = \lambda /v$ c. $C = \lambda - v$	A															
1 5	if the body is ideal black body , so its dominant color for the radiation that radiate from it is changes according to	a. the mass of the body. b. The density of the body. c. The temperature of the body. d. The type of the material that make the body.	C															
1 6	In the plank's curve for the black body radiation, if the temperature decreased	<table><tr><td></td><td>The peak of the curve</td><td>The space below the curve</td></tr><tr><td>A</td><td>Displaced to the high frequency region</td><td>Increase</td></tr><tr><td>B</td><td>Displaced to the high frequency region</td><td>Decrease</td></tr><tr><td>C</td><td>Displaced to the low frequency region</td><td>Increase</td></tr><tr><td>D</td><td>Displaced to the low frequency region</td><td>Decrease</td></tr></table>		The peak of the curve	The space below the curve	A	Displaced to the high frequency region	Increase	B	Displaced to the high frequency region	Decrease	C	Displaced to the low frequency region	Increase	D	Displaced to the low frequency region	Decrease	C
	The peak of the curve	The space below the curve																
A	Displaced to the high frequency region	Increase																
B	Displaced to the high frequency region	Decrease																
C	Displaced to the low frequency region	Increase																
D	Displaced to the low frequency region	Decrease																
1 7	In the Planck's curve for the black body radiation, the intensity of radiation at the very high frequencies is	a. Doesn't change. b. Moderate. c. Very high. d. Close to zero.	C															

1 8	At the opposite graph, if λ_1 is the shortest wavelength of the visible rays, while λ_2 is the longest wavelength for visible rays, so the graph might represent the radiation that radiate from Intensity of radiation 	a. A glowing star. b. Earth. c. Tungsten lamp. d. Human body. A
1 9	The opposite graph represents the relation between the intensity of radiation and the wavelength for two hot bodies A,B, so the ratio between their temperatures ($\frac{T_A}{T_B}$) is Intensity of radiation 	a. $\frac{4}{2}$ b. $\frac{1}{2}$ c. $\frac{1}{2}$ d. $\frac{1}{4}$ B

20	Intensity of radiation  The opposite graph shows a hot black body at T_1, T_2 so the ratio between $(\frac{\lambda_1}{\lambda_2})$ is	a. $\frac{1}{16}$ b. $\frac{1}{8}$ c. $\frac{1}{4}$ d. $\frac{1}{2}$	C
21	The opposite figure shows a hot iron piece, at which point that has the lowest temperature..... 	a. A b. B c. C d. D	A
22	A black body radiation with 3000 K, and its maximum wavelength λ, if the body is cooled to be T, its maximum wavelength become 10λ, so T equals	a. 300 K. b. 2700 K. c. 270 K. d. 1800 K.	A
23	In Planck's curve, the wavelength associating the maximum intensity of radiation emitted from the sun lies in region of	a. UV b. V.R c. I.R d. X-rays	B
24	The frequency of photons which emitted from a glowing body increases, then its number	a. Increases b. Decreases c. Remains constant d. Equals zero	B

2 5	Which of the following curves represents the classical explanation for blackbody radiation, and which one is the modern explanation 	a. A: modern B: classical. b. A: classical B: classical. c. A: modern B: modern. d. A: classical B: modern .	A
2 6	The distribution of the radiation intensity with wavelength is 	a. Wein's distribution. b. Planck's distribution. c. Classical physics distribution. d. Modern physics distribution.	B
2 7	The wavelength at maximum radiation intensity is inversely proportional to the temperature is	a. Blackbody radiation. b. Wein's law. c. Planck's law.	B
2 8	For a black body radiation if $\lambda_{\max}$ represents the maximum wavelength corresponding to maximum radiation at temperature T, the curve between $\lambda_{\max}$ and T is 	a. A b. B c. C d. D	d

2- Mention the factor affecting on

No.	Question	Answer
1	Energy of the photon	Frequency (directly) Wavelength (inversely)
2	The color of the hot objects	Temperature
3	The intensity of the black body radiation	Wavelength (inversely to a specific value then become directly) Frequency (directly to a specific value then become inversely)
4	The wavelength at maximum radiation intensity (λ_{max})	Temperature (inversely)
5	The speed of electromagnetic waves in air	It's constant (it is equal to the speed of light)
6	The speed of electromagnetic waves in media	The nature of the medium

3- True or False

No.	Question	Answer	Correction if F
1	when a piece of iron is heated to become self-luminous, the apparent color of the emitted radiation shifts initially from red to orange to yellow to blue and finally to white as the temperature increases	T	
2	The sun is yellow because its surface temperature is about 6000 K. Some stars glow blue-white because their surface temperature can be as high as 20000K	T	
3	Wave particle duality states that a wave can only act as a wave and not a particle.	F	The wave behave as a particle
4	Visible light is the part that can be seen with the human eye.	T	
5	Visible light is only a very large part of the electromagnetic spectrum.	F	Small part

6	Black body reflects no radiation and appears perfectly black.	T	
7	In Wien's law, the smaller temperature, the smaller the wave length of the peak	F	the higher temperature
8	All objects radiate energy continuously in the form of electromagnetic waves	T	

4- Mention the scientific term

No.	Question	Answer
1	It is the branch of physics that explains everything in our daily life and our common experiences	Classical physics
2	It is the branch of physics that deals with scientific phenomena in the universe which might not be directly observed in our daily life, and classical physics cannot explain	Modern (quantum) physics
3	They are vibrating electric and magnetic fields that moves through space or any other medium	Electromagnetic spectrum
4	The wavelength at maximum radiation intensity (λ_{max}) is inversely proportional to the temperature	Wien's Law
5	The wavelength (λ_{max}) at which the peak of the curve occurs is inversely proportional to temperature	Wien's Law
6	It is an ideal system that absorbs all radiation (100% of the radiation that hits it) falling on it. Then re-emit this radiation totally again	Black body radiation

7	The distribution of the radiation intensity with wavelength	Planck's distribution
8	A body that absorbs all the radiation falling on it with different wavelengths the emits them again in an ideal form .	Black body
9	The transfer of heat by electromagnetic waves such as visible light, infrared, and ultraviolet radiation.	Radiation
10	the ratio between the photon energy to its frequency.	Planck's constant (h)
11	A phenomenon used for criminology and night vision	Thermal imaging

5- complete the following sentences:

No.	Question	Answer
1	is the transfer of heat by electromagnetic waves such as visible light, infrared, and ultraviolet radiation.	Radiation
2	 is the physics that explains everything in our daily life and our common experiences.	Classical physics
3	is resulting from vibrating electric and magnetic fields that moves through space or any other medium.	Electromagnetic waves
4	 is the part that can be seen with the human eye.	Visible rays
5	There are two types of body according to its radiation,,and	Glowing – non-glowing (opaque)
6	is that deals with scientific phenomena in the universe which might not be directly observed in our daily life	Modern (quantum) physics
7	The wave particle duality means that the wave	Wave - particle

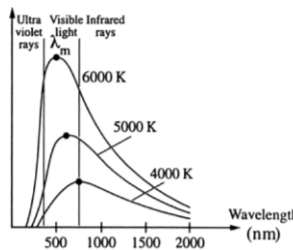
	can behave as a ,and	
8	All objects radiate energy continuously in the form of produced by thermal vibrations of the molecules.	Electromagnetic waves
9	 deal with atomic and subatomic systems.	Modern (quantum) physics
10	All phenomena involved in electronics which is the basis for all modern electronic and communication systems is explained by	Modern (quantum) physics
11	its wavelength λ ranges from 400nm to 700nm.	Visible rays
12	Above 400nm is called, while below 700nm is called	Ultraviolet rays – infrared rays
13	According to $E=h\nu$, E is Proportional with ν .	Directly proportional
14	The atom release gained energy in the form of when it return into its original orbit .	Quanta
15	Radiation is composed of small particles called	Photons
16	40% of The sun radiation, and 20% of The lamp radiation is	Visible rays
17	50% of The sun radiation, and 80% of The lamp radiation is	Heat radiation
18	The maximum intensity of radiation increases by increasing According to Planck's Distribution.	Frequency
19	The wave length at maximum radiation intensity (λ_{max}) is..... proportional to the temperature.	Inversely
20	 is an ideal system that absorbs all radiation (100% of the radiation that hits it) falling on it. Then re-emit this radiation totally again	Black body
21	is a Perfect	Black body

	absorber and perfect emitter.	
22	Classical explanation for black body radiation states that The intensity of radiation increases with the increasing of	Frequency
23	Major disagreement exists between classical theory and modern physics at	High frequency region (short wavelength)

6- Solve the following:

No.	Question	Answer
1	If the temperature of surface of sun is 6000 K and wavelength (λ_{max}) at maximum radiation intensity is 0.499 nm. Calculate the temperature of filament lamp that gives photons at wavelength (λ_{max}) = 1.2 nm	$\frac{\lambda_1}{\lambda_2} = \frac{T_2}{T_1} = \frac{0.499 \times 10^{-9}}{\frac{1.2 \times 10^{-9}}{T_2}}$ $= \frac{6000}{1.2 \times 10^{-9} \times 0.499 \times 10^{-9}}$ $= 2495K$
2	if the wavelength having a maximum intensity in the solar radiation is 0.5 μm Calculate the wavelength having a maximum intensity in the radiation emitted by Earth, given that the surface temperature of the sun and Earth are 6000 K and 300 K respectively.	$\frac{\lambda_1}{\lambda_2} = \frac{T_2}{T_1} = \frac{0.5 \times 10^{-6}}{\frac{\lambda_2}{300}}$ $= \frac{300}{6000} \times 0.5 \times 10^{-6}$ Type equation here. $= \frac{0.5 \times 10^{-6} \times 6000}{300}$ $= 10 \times 10^{-5}m$
3	calculate the wavelength of photon with energy = 4.968×10^{-19} Joule	$E = h\nu = h \frac{c}{\lambda}$ $\lambda = h \frac{c}{E}$ $\lambda = 4 \times 10^{-6}m$
4	A body at temperature 1500 K radiates out maximum energy at the wavelength 20000 Å. If the sun radiates out maximum energy at 5000 Å,	$\frac{\lambda_1}{\lambda_2} = \frac{T_2}{T_1}$

	calculate the temperature of the sun	$= \frac{20000}{5000} = \frac{T_2}{1500}$ $\frac{20000 \times 1500}{5000}$ $= 6000K$
5	What is the energy of a photon with frequency $3.0 \times 10^{14} \text{ Hz}$	$E = h\nu = (6.62 \times 10^{-34}) \times 3.0 \times 10^{14} = 1.99 \times 10^{-19} \text{ joules}$
6	The wavelength of yellow light is approximately 5.9×10^{-7} meters. What is its frequency	$C = \lambda\nu$ $\nu = \frac{C}{\lambda} = \frac{3 \times 10^8}{5.9 \times 10^{-7}} = 5.1 \times 10^{14} \text{ Hz}$
7	(a) What are the names of these curves? (b) Does the classical physics succeeded to explain these curves? And why? (c) What is meant by (λ_m)? (d) What will happen to (λ_m) when the temperature decreases?	(a) Planck's distribution (b) No, because at high frequencies the intensity of radiation decreases. (c) The maximum wavelength associating the maximum radiation intensity. (d) The maximum wavelength increases (inversely proportional)



7- Compare between the following:

1-

In terms of	Emitted rays from the sun	Emitted rays from the Earth
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Region of maximum radiation	Visible ray	Infrared radiation
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2-

In terms of	Emitted rays from the sun	Emitted rays from the electric lamp
Temperature	(T = 6000 K)	(T= 3000K)
The wavelength at the peak	$\lambda \text{ max} = 5000 \text{ \AA}$ (500 nm) (0.5 μm) (within the visible region)	$\lambda \text{ max} = 10000 \text{ \AA}$ (1000nm) (1 μm) (within the infra-red region)
the emitted energy	40% is a visible light 50% is a heat radiation 10% is a different region	20% visible light 80% heat radiation

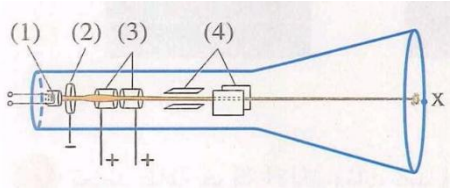
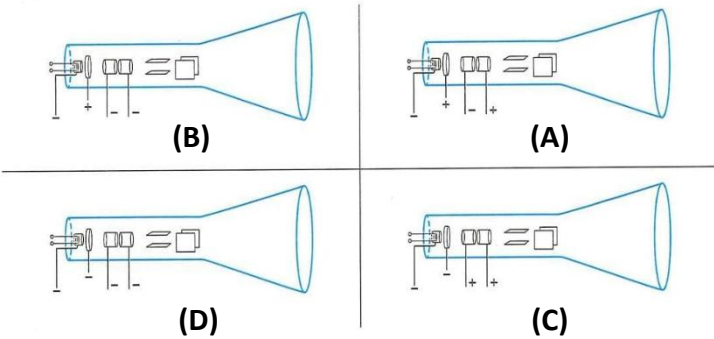
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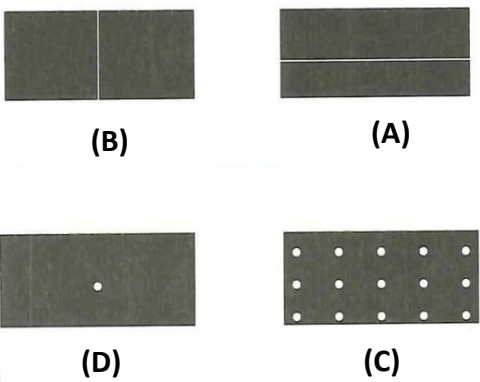
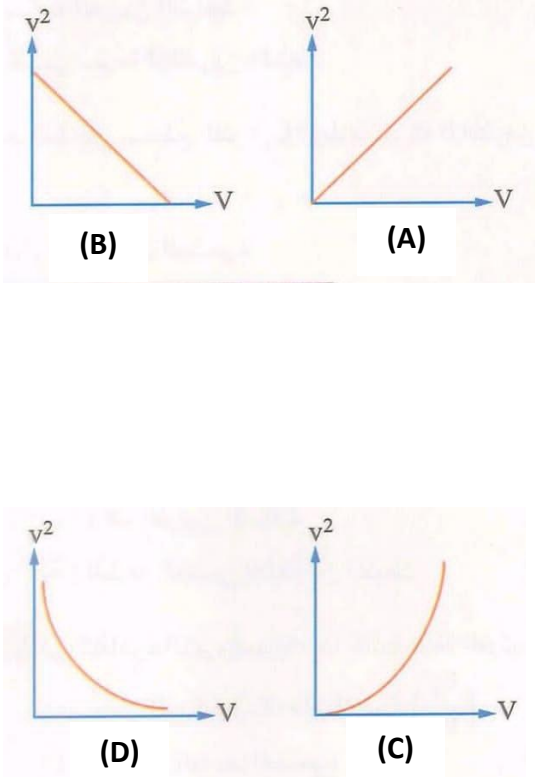
In terms of	Classical physics	Modern physics
Definition	It is that explains everything in our daily life and our common experiences.	It is that deals with scientific phenomena in the universe which might not be directly observed in our daily life.

المعرفة	الكود
Thermionic emission	TPK01

1- Choose the correct answer:

#	Question	Choices	Answer
1	The cathode rays consist of	a. Electrons b. Photons c. Electromagnetic Wave d. Gamma	a
2	Cathode rays have the energy of	a. $\frac{1}{2} m_e V^2$ b. $m_e V$ c. $2 m_e V$	a
3	The process in which electrons are emitted from a metallic surface when heat given to it is called	a. Photoelectric effect b. Thermionic effect c. Compton effect d. Black body radiation	b
4	CRT stands for	a. Common Ray Tube b. Common Running Tube c. Cathode Ran Tube d. Cathode Ray Tube	d
5	In the C.R.T when the grid voltage changes from -20 V to -50 V.....	a. Intensity of illumination increases on fluorescent screen. b. Intensity of illumination decreases on fluorescent screen. c. The deflection of electron beam increases. d. The deflection of electron beam decreases.	b
6	From the properties of beam of C.R.T.....	a. Electromagnetic waves. b. With constant speed. c. Charged particles. d. Uncharged particles.	b
7	The opposite figure represents C.R.T, which of the following parts represents..... A. The source of electrons..... B. The part that responsible for	(A): a. Part (1). b. Part (2). c. Part (3). d. Part (4). (B): a. Part (1).	(A): a (B): d (C): b

	changing the position of collision of electron beam with fluorescent screen..... C. The illumination intensity is controlled by At point X. 	b. Part (2). c. Part (3). d. Part (4). (C): a. Part (1). b. Part (2). c. Part (3). d. Part (4).	
8	Which of the following figures represents the correct voltage that is used in C.R.T.....		
9	In the C.R.T when the filament is damaged	a. Intensity of illumination increases on fluorescent screen. b. Intensity of illumination decreases on fluorescent screen. c. The fluorescent screen doesn't illuminate. d. The deflection of electron beam decreases.	c
10	In the C.R.T when the grid is not connected to any electric signal	a. It is hard to control the path of electric beam into the screen. b. The fluorescent screen doesn't illuminate. c. The electric beam reflects to cathode. d. The illumination remains constant.	d

1 1	Which of the following figures represents the final image that formed in the screen of C.R.T when the deflection system doesn't work.....	 (B) (A) (D) (C)	d
1 2	The kinetic energy for the electrons is depend on when it reaches the Anode in C.R.T.	a. The surface area of cathode. b. Work done for Anode. c. Potential difference between two plates of deflection system. d. Potential difference between cathode and anode.	c
1 3	In C.R.T the velocity of electron is v , when the electron is accelerated by potential difference with V , if the potential difference is increased to $2V$, son the velocity will become.....	a. $2 v$ b. $\sqrt{4 v}$ c. $4 v$ d. $1/2 v$	c
1 4	Which of the following graphs represents the relationship between v^2 for the electron and V in C.R.T.....	 (B) (A) (D) (C)	a

1 5	If the potential difference between anode and cathode in C.R.T is 1000 V : so...‘ A. K.E for the electron is B. The maximum velocity of liberated electron from cathode when it reaches anode equals.....	(A): a. $1.6 \times 10^{-15} J$ b. $1.6 \times 10^{-16} J$ c. $1.6 \times 10^{-17} J$ d. $32 \times 10^{-18} J$ (B) : a. $1.88 \times 10^6 m/s$ b. $1.88 \times 10^7 m/s$ c. $3.52 \times 10^5 m/s$ d. $3.52 \times 10^4 m/s$	(A) b (B) b
1 6	The source of thermal energy in the C.R.T is.....	a. Anode. b. Cathode. c. Filament. d. Fluorescent screen.	c
1 7	The reason for the kinetic energy of the electron is.....	a. Velocity. b. Density c. Mass of the electron. d. Potential difference between cathode and anode.	d
1 8	It is used to accelerate the electron beam towards the screen.....	a. Cathode. b. Anode. c. Grid. d. Filament.	b
1 9	To increase to velocity of electron in C.R.T for four times.....	a. Increase K.E for four times. b. Increase the mass of electron to double. c. Increase the potential difference to double. d. Increase the potential difference for four times.	c
2 0	To decrease the intensity of electron beam	a. Increase the voltage of grid. b. Decease the voltage of anode. c. Increase the voltage of anode. d. Decrease the voltage of grid.	a
2 1	Which part of C.R.T is responsible for the deflection of electron beam.....	a. Vertical plates. b. Horizontal plates. c. Anode. d. A,B.	d

2- Mention the application of

No	Question	Answer
1	The electron gun in CRT	It is an instrument to produce a beam of electrons at high speed
2	CRT	used in TV and computer monitors
3	Deflection system in CRT	Deflect the path of the electron
4	fluorescent screen in CRT	Emits green light when the electron strikes it
5	Filament in CRT	It is used to heat up the cathode
6	Cathode in CRT	Emission of electrons
7	Grid in CRT	Control the number of electrons in the electron beam
8	Anode in CRT	It is used to accelerate and focus the electron beam towards the screen

3- Mention the factor affecting on

No.	Question	Answer
1	KE of the freed electron in CRT	Accelerating voltage or velocity of the electron
2	the velocity in freed electron in CRT	Accelerating voltage or kinetic energy

4- Solve the following:

No.	Question	Answer
1	An electron moving at a constant speed 4.4×10^5 m/sec, what is its kinetic	$KE = \frac{1}{2} m_e v^2$

	energy?	$\frac{1}{2} m_e (4.4 * 10^5)^2$ $8.82 * 10^{-20} J$
2	Calculate the speed of an electron that has kinetic energy of 120 eV.	$120 * e = \frac{1}{2} m_e v^2$ $v = 6497051 \text{ m/s}$
3	calculate the potential difference which accelerated an electron from rest with speed equals $9.796 * 10^6 \text{ m/s}$	$\frac{1}{2} m_e v^2$ $V = \frac{\quad}{e}$ $V = 272.8 \text{ volts}$

5- complete the following sentences:

No.	Question	Answer
1	There are many ways to eject (escape) electrons out of a materialand	Thermionic emission – photoelectric effect
2	Emission of electrons from a heated metal's surface (by heating) is called	Thermionic emission
3	 is application on the thermionic emission.	Cathode ray tube (CRT)
4	C.R.T is used to make and Monitors.	TV – computer
5	 is the attractive force between the positive ions and free electrons in metal which make the electrons cannot leave it.	Surface potential barrier
6	Operation stages for C.R.T areand	Electron gun – deflection system
7	 is used to control the number of electrons of the electron beam .	grid
8	 is used to accelerate the electron beam towards the screen.	Anode
9	The stage at which the beam at high speed.....	Electron gun

10	emits green light when the electron strike it.	Fluorescent screen
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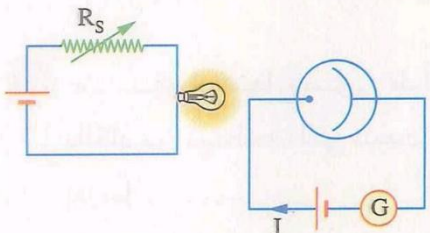
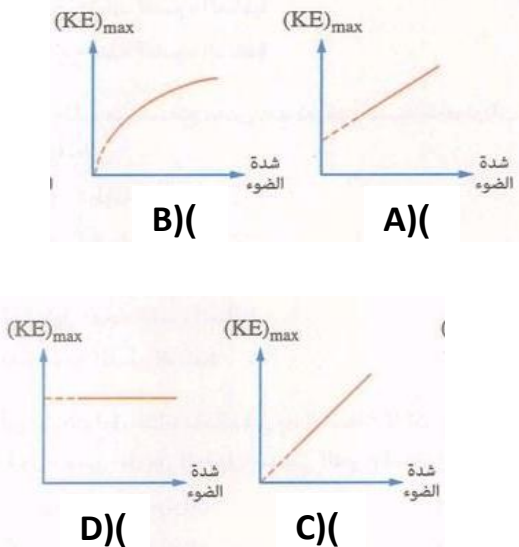
6- Mention the scientific term

No.	Question	Answer
1	emission of electrons from a heated metal's surface (by heating)	Thermionic emission
2	emission of electrons from a metallic surface when the light energy falling on it	photoelectric effect
3	It is the attractive force between the positive ions and free electrons in metal which make the electrons cannot leave it	Surface potential barrier

المعرفة	الكود
Photoelectric effect	TPK01

1-Choose the correct answer:

#	Question	Choices	Answer
1	The process in which electrons are emitted from a metallic surface when a light given to it is called	a. Photoelectric effect b. Thermionic effect c. Compton effect d. Black body radiation	a
2	the emission of electrons from metallic surface (photoelectric effect) depends on	a. The intensity of the incident light b. The frequency of incident light c. The exposure time	b
3	The process in which electrons are emitted from a metallic surface when a light given to it is called	a. Photoelectric effect b. Thermionic effect c. Compton effect d. Black body radiation	a

4	The emission of electrons from cathode in photoelectric cell is depend on	a. The type of material of Anode and the light intensity. b. The type of material of Anode and the light frequency. c. The type of material of Cathode and the light intensity. d. The type of material of Cathode and the light frequency.	d
5	At the opposite figure shows light produced from an electric lamp falls on a photoelectric cell cases flow of photocurrent, if the light intensity increases, the photocurrent 	a. Increases. b. Decreases. c. Vanishes. d. Doesn't change.	d
6	Which of the following graphs represents the relationship between $K.E_{Max}$ for emitted photoelectrons and light intensity	 A)(B)(C)(D)(	d

7	An electromagnetic radiation is incident on a photoelectric cell with ν , photoelectrons emitted with $K.E_{Max}$, if the frequency becomes 2ν , which of the following doesn't change	a. The photon energy. b. Speed of photon c. $K.E_{Max}$ for emitted photoelectron. d. Maximum velocity of emitted photoelectron.	b
8	A monochromatic light incident on photoelectric cell with frequency greater than the critical frequency for cathode, then another light is used with the same wavelength of the first one but its intensity is double of the first, which one is doubled	a. The energy of each photon e. $K.E_{Max}$ for emitted photoelectron. b. Work function for the metal. c. The intensity of photocurrent.	C Because the frequency of incident light greater than E_w , so the intensity of photoelectrons is directly with the light intensity.
9	If the frequency of incident light increase to double, so the work function	a. Increase to double b. Decrease to half c. Remains constant d. Increase for four times.	c
10	When the incident light has wavelength 623 nm on a metal, photoelectrons are emitted with 4.6×10^5 m/s, so : A- The work function of the metal B- The critical frequency of the metal	(A) a. $2.23 \times 10^{-19} J$. b. $3.19 \times 10^{-19} J$. c. $4.15 \times 10^{-19} J$. d. $4.6 \times 10^{-19} J$. (B) a. 6.94×10^{14} b. $6.26 \times 10^{14} HZ$. c. $4.81 \times 10^{14} HZ$. d. $3.34 \times 10^{14} HZ$.	(A) b (B) c
11	If the critical frequency of a metal equals the frequency of the blue light, which of the following rays will be able to emit the electrons from the metal surface.....	a. Infrared rays. b. Radio waves. c. Red light. d. Ultraviolet rays.	d

12	If a light with $3100 \text{ \AA}$ incident on the surface of cathode in photoelectric cell, so the electrons are emitted with kinetic energy equals 2.5 eV , so the work function for the cathode is	a. 3.1 eV . b. 2.4 eV . c. 1.5 eV . d. 0.9 eV .	b
13	If the work function for a metal equals $3.3125 \times 10^{-19} \text{ J}$, the critical frequency is	a. $4.5 \times 10^{14} \text{ HZ}$. b. $4.8 \times 10^{14} \text{ HZ}$. c. $5 \times 10^{14} \text{ HZ}$. d. $5.5 \times 10^{14} \text{ HZ}$.	c

2- Mention the factor affecting on

No.	Question	Answer
1	KE of the freed electron by photoelectric effect	Incident photon energy or Work function or frequency of light used
2	the velocity of the freed electron by photoelectric effect	Kinetic energy

3- True or False

No.	Question	Answer	Correction if F
1	Changing the color of the light means we are changing the frequency.	T	
2	In Photoelectric effect, electrical energy converted to light energy	F	Light to electrical
3	Photoelectrons are electrons that are released or ejected from a substance by photoelectric effect	T	
4	Kinetic energy is independent of the intensity of light.	T	

5	Increasing the brightness of the light, increases the speed of the electrons.	F	independent
6	Increasing the intensity of the light, increases the number of electrons.	T	
7	increasing the frequency, increases the kinetic energy.	T	
8	Photoelectric effect is always instantaneous	T	
9	The photon energy doesn't have to be more than the work function for electrons to be ejected.	F	It has to be greater

4-Solve the following:

No.	Question	Answer
1	a monochromatic radiation of wavelength 3×10^{-7} m falls on a metal surface, so electrons liberated with maximum kinetic energy 3.26×10^{-19} J a) the energy of the photon that falls on the metal surface b) the work function of the metal	a) Photon energy = $h\nu = h \cdot \frac{c}{\lambda}$ 6.63×10^{-19} J b) $E_w = h\nu - KE$ 3.37×10^{-19} J
2	photon with wavelength 4×10^{-7} m falls on a surface of metal with work function 2.3×10^{-19} J, find the kinetic energy of emitted electron from a surface of the metal	$KE = h \cdot \frac{3 \times 10^8}{4 \times 10^{-7}} - 2.3 \times 10^{-19}$ $2.67 \times 10^{-19} \text{ J}$
3	A metal has a work function 8×10^{-19} J. Calculate the maximum kinetic energy of electrons emitted from the metal when light of frequency 2×10^{15} Hz is shone on the surface.	
4	When monochlight of wave length 663 nm is incident on a metal plate whose work function 3.52×10^{-19} eV will electrons be emitted or not?	
5	Draw the relationship between light intensity and photocurrent at the case of $E < E_w$ and $E \geq E_w$.	(a) $E < E_w$ (b) $E \geq E_w$

5- complete the following sentences:

No.	Question	Answer
1	There are many ways to eject (escape) electrons out of a material are and	Thermionic emission – photoelectric effect
2	The emission of electrons from a metallic surface when the light energy falling on is	Photoelectric effect
3	The electrons that released or ejected from a substance by photoelectric effect is called	Photoelectrons
4	An application on photoelectric effect is	Photoelectric cell
5	photoelectric cell aims to convert To	Light energy-electric energy
6	the source of free electrons in photoelectric cell is	Cathode
7	the collector in photoelectric cell is called	Anode
8	the minimum energy needed to free (release) the electrons from the surface of the metal is	Work function
z	To free (emit) an electron with kinetic energy from the metal, the energy of the incident photon must be.....	Electron gun
10	The kinetic energy of the emitted electrons is independent of	Fluorescent screen
12	At the case of the relationship between light intensity and photocurrent is independent of each other.	$E < E_w$
13	At the case of the relationship between light intensity and photocurrent is directly proportional to each other.	$E > E_w$

5- Mention the scientific term

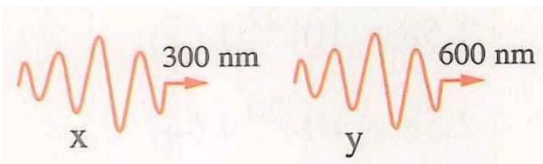
No.	Question	Answer
1	An electron released or ejected from a substance by photoelectric effect	Photoelectrons
2	It is the minimum energy needed to free (release) the electrons from the surface of the metal	work function of the metal

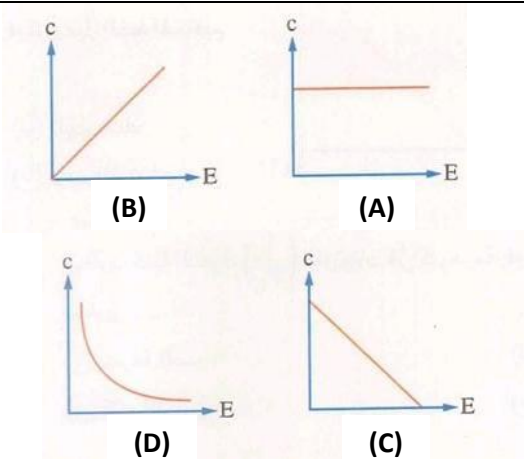
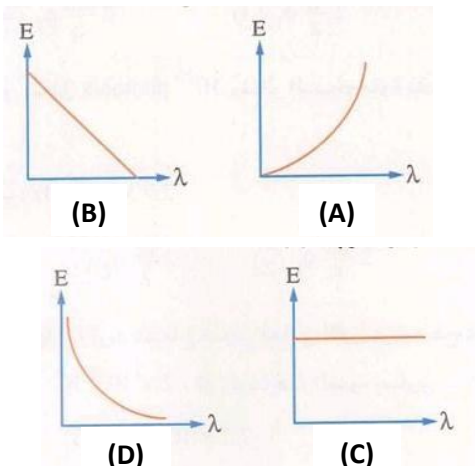
المعرفة	الكود
Compton Scattering	TPK02

1-Choose the correct answer:

#	Question	Choices	Answer
1	The ratio between the energy of the photon to the velocity of light in air is the of the photon	a. Mass b. Frequency c. Kinetic energy d. Momentum	d
2	the ratio of the photon's momentum to its mass equals	a. The velocity of light b. Planck's constant c. The photon energy d. photons are massless particles	a
3	in Compton effect, ratio between the photon energy after collision to its energy before collision is	a. Greater than one b. Equal one c. Less than one	c
4	In Compton effect, the ratio between the velocity of photon after collision to the velocity of photon before collision	a. Greater than one b. Equal one c. Less than one	b
5	In Compton effect, the ratio between the frequency of photon after collision to the frequency of photon before collision	a. Greater than one b. Equal one c. Less than one	c

6	In Compton effect, the ratio between the wavelength of photon after collision to the wavelength of photon before collision	a. Greater than one b. Equal one c. Less than one	a
7	in Compton effect, the speed of gamma rays photons after collides with a free electron is	a. Increased b. Decreased c. Equal zero d. Does not change	d
8	in Compton effect, gamma photons is scattered and its increases	a. Energy b. Speed c. Wavelength d. Momentum	c
9	When a photon of high frequency (x- rays or gamma rays) collides with a free electron, the photon frequency decreased and changed its direction; also the electrons velocity increased and changed its direction	a. Characteristic radiation b. Continuous radiation c. Compton effect d. Thermionic emission	c
10	The Compton effect proves	a. The particle nature of photon. b. The wave nature of photon. c. The wave nature of electron. d. The particle nature of electron .	a
11	In Compton effect, when a photon with high frequency collides with free electron , which of the following will increase for photon after collision	a. Energy . b. Frequency. c. Wavelength. d. Linear momentum.	c
12	In Compton effect after collision , there is no decrease of	a. The wavelength of electron. b. Energy of photon. c. Photon frequency. d. Photon speed.	d
13	From the properties of photon	a. Its speed equals the speed of light . b. Deflects by the electric field. c. It may be accelerated. d. All the above.	a

1 4	At the opposite figure, the ratio between the speed of photon x and y is..... 	a. $\frac{4}{1}$ b. $\frac{2}{3}$ c. $\frac{1}{1}$ d. $\frac{1}{3}$	c
1 5	The linear momentum for a photon is	a. $\frac{h}{\lambda}$ b. $\frac{h\nu}{\lambda}$ c. $\frac{hc}{\lambda}$ d. $\frac{h\nu}{c^2}$	a
1 6	Photon x has P_L while photon y has $2 P_L$, so	a. The wavelength of x is double wavelength of y. b. The wavelength of x is half wavelength of y. c. The speed of x is double speed of y. d. The speed of x is half speed of y.	b
1 7	Two photon the ratio between their frequencies is 1:2, so the ratio between their energies is	a. 1:1 b. 1:2 c. 2:1 d. 4:1	b
1 8	A monochromatic light its number of photons is n and its wavelength is λ , so the total energy is	a. $\frac{hc}{n\lambda}$ b. $\frac{n\lambda}{hc}$ c. $\frac{nhc}{\lambda}$ d. $\frac{nc}{\lambda}$	c

19	Which of the following graphs represents the relation between the speed of photon and its energy		a
20	Which of the following graphs represents the relation between the energy of photon and its wavelength.....		d
21	Photons are electrically neutral particle and they are not deflected by the either electric or magnetic field.....	a. True b. False c. Non of the above	a
22	1. According to Compton effect when a photon collides with a free electron. Then,	a. the photon loses its energy. b. the electron gains a kinetic energy equals that of the photon energy. c. the frequency of the scattered photon is less than the frequency of the incident photon. d. the electron and the photon move together on the same line.	c
23	In Compton experiment , the sum of the photon energy and the electron energy before collision is the sum of their energies after collision.	a. greater than b. less than c. equal to d. no correct answer	c

2 4	In Compton effect, the ratio between the velocity of photon after collision to the velocity of photon before collision.....	a. Greater than one b. Equal one c. Less than one	b
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2- True or False

No.	Question	Answer	Correction if F
1	When a photon of high frequency (x- rays or gamma rays) collides with a free electron, The photon's frequency decreases	T	
2	When a photon of high frequency (x- rays or gamma rays) collides with a free electron, The photon's wavelength decreases	F	increase
3	When a photon of high frequency (x- rays or gamma rays) collides with a free electron, the photon's speed decreases.	F	Stays the same
4	When a photon of high frequency (x- rays or gamma rays) collides with a free electron, the photon's speed increases.	F	Stays the same
5	When a photon of high frequency (x- rays or gamma rays) collides with a free electron, the photon's speed remains the same.	T	
6	When a photon of high frequency (x- rays or gamma rays) collides with a free electron, the electron's speed increases.	T	
7	When a photon of high frequency (x- rays or gamma rays) collides with a free electron, the electron's speed decreases.	F	
8	Classical theories can explain the change in wavelength.	F	Can't
9	The conservation of energy states that energy of the photon before and after collision is the same.	F	
10	The conservation of energy states that energy of the electron before and after collision is the same.	F	
11	The conservation of energy states that energy of the sum of energy of photon and electron before and after collision is the same.	T	

3-Solve the following:

No.	Question	Answer
1	A photon of X rays of ($\lambda = 3 \text{ \AA}$) collides a stationary electron, the electron is scattered with energy $1.1 \times 10^{-16} \text{ J}$, so the wavelength of the scattered photon is $\text{\AA}$	
2	calculate the photon frequency, mass and linear momentum if $\lambda = 380 \text{ nm}$	
3	the energy of photon $3.3 \times 10^{-19} \text{ joule}$. Calculate its mass, frequency and wavelength.	a. photon is massless b. $f = E/h$ $f = 4.98 \times 10^{14} \text{ Hz}$ c. wavelength = c/f $6.02 \times 10^{-7} \text{ m}$
	From studying Compton effect, mention what happens after collision for each value of the following, then mention the reason : (1st session 10) (a) The photon energy. (b) The photon velocity.	(c) The energy of photon after collision is decreased. (d) The photon velocity after collision remains constant.

4- Mention the scientific term

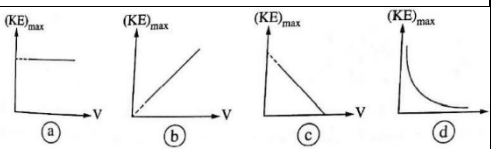
No.	Question	Answer
1	The total linear momentum is the same before and after collision	The conservation of momentum
2	The total energy is the same before and after collision	The conservation of energy

المعرفة	الكود
X-Ray	TPK02

1-Choose the correct answer:

#	Question	Choices	Answer
1	The wavelength of x-rays range is ... in Å	a. 10^{-13} to 10^{-8} b. 10^{-13} to 10^2 c. 10^{-3} to 10^2 d. 10^{-3} to 10^{-8}	c
2	The wavelength of x-rays range is ... in meters	a. 10^{-13} to 10^{-8} b. 10^{-13} to 10^2 c. 10^{-3} to 10^2 d. 10^{-3} to 10^{-8}	a
3	This radiation is called soft X-rays	a. Characteristic radiation b. Continuous radiation c. Both a and b	b
4	This radiation is sometimes called bremsstrahlung	a. Characteristic radiation b. Continuous radiation c. Both a and b	b
5	Bremsstrahlung is a ... word	a. Latin b. Swedin c. German	c
6	Bremsstrahlung means	a. braking radiation b. breaking radiation c. barking radiation	a
7	The x-ray is considered	a. Visible electromagnetic wave. b. Invisible electromagnetic wave. c. Mechanical wave. d. UV ray.	b

8	The tube that is used to produce x-ray	a. C.R.T b. Photocell c. Coolidge tube d. Resonant cavity.	c
9	The filament in the Coolidge tube is responsible for.....	a. Source of heat energy. b. Source of free electrons. c. A,B. d. None of these.	c
10	The cooling fins are responsible for.....	a. Reduce light energy. b. Reduce x-rays. c. Reduce heat energy. d. Reduce electric energy.	c
11	The spectrum of the x-rays is	a. Continuous spectrum. b. Line spectrum. c. Both .	c
12	The line spectrum resulted from	a. Characteristic radiation. b. Thermal radiation. c. Soft radiation. d. Braking radiation.	a
13	The continuous spectrum resulted from....	a. Characteristic radiation. b. Thermal radiation. c. Black body radiation. d. Braking radiation.	d
14	The Characteristic radiation is depend on	a. the kind of target element. b. the voltage difference. c. None of these	a
15	The continuous radiation is depend on	a. the kind of target element. b. the voltage difference. c. None of these.	b
16	Which of the following graphs represents the relation between the maximum frequency (ν_0) of the photons of the continuous spectrum of the X-rays which are generated from the Coolidge tube and the potential difference (V) between the anode and the cathode?		b

1 7	In the Coolidge tube, if the potential difference between the cathode and the anode is doubled, then the shortest wavelength for the braking spectrum of the X-rays (a) does not change (b) decreases to quarter its initial value (c) decreases to half its initial value (d) is doubled	d
1 8	Which of the following graphs represents the relation between the maximum kinetic energy ($(KE)_{\max}$) acquired by one of the electrons which is emitted from the cathode in the Coolidge tube and the potential difference (V) between the cathode and the anode? 	b

2- True or False

No.	Question	Answer	Correction if F
1	The continuous spectrum produces all wavelengths within a certain range regardless of the target material	T	
2	Continuous radiation is generated when an electron collides with an electron close to the nucleus of the target material atom.	F	Characteristic radiation
3	The wavelength of the characteristic radiation (at the peak) does not depend on the voltage difference	T	
4	The wavelength of the characteristic radiation (at the peak) does not depend on the kind of target element	F	Depends on it
5	The wavelength of the characteristic radiation (at the peak) depends on the voltage difference	F	Doesn't depend
6	The higher the atomic number of the target element, the shorter the wavelength of the characteristic radiation	T	
7	At low voltage differences, no characteristic radiation is produced.	T	
8	The characteristic radiation is generated when the velocity of the colliding electrons is reduced upon passing through a retarding electric field due to the electrons of the target material.	F	Continuous radiation

3-Mention the application of

No.	Question	Answer
1	Coolidge tube	Production of X-rays
2	Cooling fin in Coolidge tube	to reduce heating effect
3	vacuum tube in Coolidge tube	to avoid the collision with air molecules
4	An external high DC voltage source in an external high DC voltage source	Accelerate electrons towards the target
5	Filament in Coolidge tube	Increase the temperature of cathode

4- Mention the factor affecting on

No.	Question	Answer
1	the minimum wavelength of x-rays of continuous radiation	Potential difference between cathode and anode
2	characteristic x-rays radiation	The difference of energy between k and l levels in target material

5-Solve the following:

No.	Question	Answer
1	X-rays emits by wavelength 632 nm, what is the frequency of x-rays in Hz	$f = \frac{c}{\lambda}$ $4.75 \times 10^{14} \text{ Hz}$
2	calculate the maximum frequency of the generated X rays from Coolidge tube when the voltage difference equal to 10000 volt	$f = \frac{e * 10000}{h}$ $2.42 \times 10^{18} \text{ Hz}$
3	X-rays are generated in the x-rays tube with operating voltage of 20kv. Calculate: a) the maximum energy of the emitted photons b) the maximum frequency of the emitted photons	a. $E = e * V = e * 20,000$ $3.2 \times 10^{-15} \text{ J}$ b. $f = \frac{e * 20000}{h}$ $4.84 \times 10^{18} \text{ Hz}$
4	State the applications of x-ray	a. Studying the crystalline structure b. Detecting the defects in metallic structure c. In the medicine

5	State the properties of X-Ray	a. They can penetrate media easily b. They can ionize gases c. They diffract in crystals d. They affect sensitive photographic plates
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المعرفة	الكود
LASER	TPK02

1-Choose the correct answer:

#	Question	Choices	Answer
1	laser beam production is based on the principle of	a. Induced absorption b. spontaneous emission c. ionization d. simulated emission	d
2	from the characteristic of laser beam	a. monochromatic b. non coherence c. varying in phase d. all of the above	a
3	laser light is monochromatic because	a. It has approximately one wavelength b. Its photons are of the same phase c. It is parallel rays d. all of the above	d
4	laser beam has high intensity, which means	a. It doesn't obey inverse square law b. It has only one wavelength c. Its photons have the same direction d. all of the above	a

5	which of the following is not true for laser	a. Extremely intense light b. Perfectly monochromatic c. Coherent d. Following inverse square law	d
6	laser beam doesn't follow inverse square law because of	a. Its monochromaticity b. Its coherent c. Its non-divergence d. All of the above	c
7	the common feature of laser and (x) rays photons are that they	a. Are coherent b. Are monochromatic c. Have the same speed d. Have the same energy	c
8	the velocity of laser beam is the velocity of ordinary light	a. Higher than b. Equal to c. Lower than	b
9	coherent photons in the optical rays means	a. Emitted by variable phase difference b. Moving in parallel beam c. Emitted by constant phase difference d. Doesn't obey inverse square law	c
10	if the intensity of a laser ray 10cm away from its source is (I), so its intensity value away from 20cm is	a. $I/4$ b. $I/2$ c. I d. $2I$	c
11	from the main components of laser	a. Electrons b. Holes c. Active medium - d.	c
12	population inversion is state of the system, where in the upper energy state compared to lower energy state is populated	a. Less than b. More than c. Equal d. None of the above	b

13	The lowest energy state is called	a. Ground states. b. Excited states. c. All the above.	a
	The highest energy state is called	a. Ground states. b. Excited states. c. All the above.	b

2-Mention the application of

No.	Question	Answer
1	active medium in laser	The effective material required to produce laser
2	Source of energy in laser	it is the responsible for exciting the active medium
3	Resonant cavity in laser	it is the container and the activating catalyst for amplification

3-Mention the scientific term

No.	Question	Answer
1	the acronym for Light Amplification by Stimulated Emission of Radiation	LASER
2	Occurs when the atom relaxes from an excited state to a lower state after lifetime interval is over (nearly 10^{-8} s) emitting spontaneously the energy difference in the form of a photon without the effect of an external photon	Spontaneous emission
3	Occurs where an external photon of energy $E_2 - E_1$ passes by excited atom at E_2 to relax it to E_1 before lifetime is over	Stimulated emission
4	the laser emits energy over a very narrow wavelength range, literally, one color.	Monochromaticity of Laser beam
5	The diameter of the emitted light beam stays constant for long distance without much un scattering Thus, energy is	collimation of Laser beam

	transmitted without much loss.	
6	the emitted Photons are in phase and in the same direction	coherence
7	Photons emanate coherently both in time and place, since they come out together at the same time sequence	Coherence of Laser beam
8	The number of atoms in the excited state exceeds the number of atoms in the lower	Population inversion
9	the Photons are increased in number going back and forth in the active medium, due to multiple reflections between two enclosing mirrors	Amplification
10	The effective material required to produce laser	Active medium
11	it is the responsible for exciting the active medium	Energy source
12	it is the container and the activating catalyst for amplification	resonant cavity
13	It is an excited energy level characterized by its lifetime is relatively long nearly 10^{-3} s	Meta stable lifetime energy level

4-Solve the following:

No.	Question	Answer
1	A laser emits light with difference in two energy levels E_2 & E_1 is 3 eV then the wavelength of the radiation in nm is (to convert eV to J , multiply eV in the charge of the electron)	